

15.

Ans: A

Let the total energy of the oscillating system be $E = \frac{1}{2}m\omega^2 x_0^2$.

Since potential energy of the system is $\frac{1}{2}m\omega^2 x^2$, when $x = \frac{1}{2}x_0$, the potential energy is $\frac{E}{4}$. Then the kinetic energy would have to be $\frac{3E}{4}$. Hence ratio is $\frac{1}{3}$

16.

Ans: C